

Murilo Soares

Full-stack Developer · Python, TypeScript and SQL

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SUMMARY

3+ years building web applications for real clients, currently studying Systems Analysis and Development. I came from the business side — websites and paid-traffic campaigns where conversion was the only metric — and brought the habit of measuring before deciding. I write Python and TypeScript with tests, model relational databases, and ship what I build: **five public repositories** with green CI, measured numbers and a written record of every decision. One is live with an open demo account.

EXPERIENCE

Convertta — Web Developer & Paid Traffic Manager

Jan 2023 – Present · freelance

- Websites and landing pages built to convert, not to decorate: HTML, CSS and JavaScript, with layout decisions justified by conversion rate and performance score, across three segments: SaaS, e-commerce and legal.
- End-to-end Google Ads and Meta Ads campaigns — funnel structure, creative, cost-per-lead tracking and optimisation by number rather than by feel. This is where the questions behind the data projects below came from.

PROJECTS

Painel Convertta — full-stack app, live

FastAPI · PostgreSQL · React · TypeScript · Docker

Open source · [painel-convertta-web.onrender.com](#) (demo account) · [github.com/SoaresM-dev/painel-convertta](#)

- The tool I needed: leads and campaigns from several clients in one place, with computed cost per lead, date ranges, channel breakdown, funnel and time series.
- 120 tests running against PostgreSQL in CI**, each with a clean database. One of them pins the fan-out bug — joining campaigns and leads in the same JOIN would multiply spend by the number of leads.
- Money in cents, always integer; the summary is **one query, not a loop** (the naive version makes 1+2N round trips). `docker compose up` brings everything up in one command, and a workflow watches production.

Aiden — fully offline voice assistant

Python · FastAPI · pytest · Ollama · WebSocket

Open source (MIT) · [github.com/SoaresM-dev/aiden](#)

- Complete voice pipeline running **100% offline** on a laptop with no dedicated GPU and 15.7 GB of RAM: locally trained wake word, faster-whisper, an intent router with 27 skills, a local LLM through Ollama, Piper and a WebSocket HUD. No audio leaves the machine and there is no API key anywhere.
- Time to first answer from **14.8s to 8.5s (-42%)** and transcription error rate from 31% to 24.7%, on the same hardware — the model cascade came out of a 25-audio bench, not a guess.
- 531 tests green in CI**. I diagnosed a suite hang caused by an audio double misaligned by a recalibration, and split the tests that depend on this machine from the ones that hold anywhere.

CNPJ Lookup — API

Node.js · TypeScript · node:test

Open source · [github.com/SoaresM-dev/consulta-cnpj](#)

- Looks up public company records, with caching, rate limiting and selective retry — it retries what is a transient failure and gives up on what will not improve. **35 tests, none of them touch the network**; two came out of the suite carrying state between cases, making results depend on execution order.

Paid-traffic data model and queries

PostgreSQL · MySQL · SQL · GitHub Actions

Open source · [github.com/SoaresM-dev/sql-trafego-pago](#)

- Star schema and ten queries that answer the questions of a real client meeting: cost per lead, ROAS, funnel, cohort, budget overrun. Window functions, PERCENTILE_CONT, NOT EXISTS and HAVING over aggregates.
- Deterministic seed: CI compares each query's output against an expected file. **3.6x gain from an index**, measured with EXPLAIN ANALYZE over one million rows.
- All ten also on **MySQL 8**, both dialects in CI: three differences change the number without raising an error, and PERCENTILE_CONT, absent in MySQL, was rebuilt with window functions.

Data structures in C

C11 · Valgrind · gcc and clang

Open source · [github.com/SoaresM-dev/estruturas-c](#)

- Dynamic array, doubly linked list and circular queue in strict C11. **23 tests, Valgrind clean**, compiling with -Werror on both compilers, and a bench that measures O(1) against O(n).

EDUCATION

Systems Analysis and Development (Technologist)

2023 – Jul 2028

IFTM · Data Structures · Databases · Web Scripting

SKILLS

Backend	Python, FastAPI, SQLAlchemy, Alembic, pytest, Node.js, REST, JWT, WebSocket
Data	PostgreSQL, MySQL, SQL (window functions, CTEs, indexing), relational modelling, SQLite
Frontend	JavaScript, TypeScript (strict mode), HTML, CSS, React, Vite, accessibility (WCAG), Lighthouse
Infrastructure	Git, GitHub Actions, Docker, Docker Compose, Render, Linux, Windows
Other	C, Google Ads, Meta Ads, Analytics, Claude Code and Codex in the daily loop
Languages	Portuguese native · English advanced — technical reading, writing and conversation